Chapter 3 Literature Review: Before and After Comparison

Report: Data Driven Prediction of Local Government Water Service Delivery Outcomes through Machine Learning (N Hlumbene)
Compared: Chapter 3 in *Final Report.docx* (before) and *Final Report (Chapter 3 revised).docx* (after)
Date of comparison: 1 November 2026

Table 1: Summary at a glance

Measure	Before	After
Body word count	2 464	4 900
Body paragraphs	25	40
Main sections	6 (3.1 to 3.6)	7 (3.1 to 3.7)
Total headings	9	16
Organisation	Topic by topic, with no stated organising approach	Thematic, stated explicitly in the introduction, following the literature review template
Distinct sources cited	16	24 (adds Obunga et al., 2025 and Phumlani et al., 2025 from elsewhere in the report, plus six new verified sources: Breiman, 2001; Chicco & Jurman, 2020; Kapoor & Narayanan, 2023; Lundberg & Lee, 2017; Saito & Rehmsmeier, 2015; Troyanskaya et al., 2001)
Distinct in text citations (author, year and page)	22	44
Sources outside the original reference list	None	Six, each verified and added to the reference list (see the separate New Sources Register PDF)
Hyphenated words and dashes	24 hyphenated words (for example data-driven, non-technical, Sub-Saharan, trade-off)	0
Spelling	Mostly British or South African; "program" used for a programme of work	South African English throughout (organise, ageing, sceptical, centred, utilisation); decimal commas (90,6%)
Heading styles	Inconsistent: 3.1 at Heading 2, but 3.2 to 3.4 at Heading 3 and their subsections at Heading 4	Consistent: all 3.x headings at Heading 2 and all 3.x.x headings at Heading 3

Table 2: Section mapping

Before	After	What changed
3.1 Introduction	3.1 Introduction	Rewritten to cover the four template points: introduce the topic, state the purpose (linked to both theoretical objectives),

		define what is included and excluded, and state that the review is thematic with a roadmap of sections.
None	3.2 Theoretical and Conceptual Background (3.2.1 to 3.2.4)	New section. Introduces the constitutional mandate, the compliance performance gap, supervised machine learning with KNN imputation and SHAP, and a conceptual framework that connects them. Content drawn from concepts and citations already in Chapter 1.
None	3.3 Review of Empirical Studies	New umbrella section that groups the three themes and explains their order.
3.3 Water Service Delivery Status and Challenges in South Africa	3.3.1 (Theme 1)	Moved to the first theme so the problem is established before its causes and solutions. Paragraphs restructured into theme sentence, authors, comparison and own voice. Ends with a link sentence to Theme 2.
3.4 The Relationship Between Clean Audit Outcomes and Actual Service Delivery, with 3.4.1 The Compliance-Performance Gap	3.3.2 (Theme 2)	Now the second theme. The Midvaal study (Khasodi et al.) is corrected to match the paper's conclusion: it agrees with Sambo and Craig that clean audits alone do not show good service delivery, and adds that internal controls drive both service capacity and clean audits. Ends with a link to Theme 3.
3.2 Machine Learning as a Predictive Tool in Public Service Delivery, with 3.2.1 Sectoral Machine Learning Applications	3.3.3 (Theme 3)	Moved from first to third theme, so machine learning is presented as the response to the problem rather than the opening topic. Comparisons between authors added (for example Taweesan et al. compared with Hirko et al.).
None	3.3.4 Data Preparation, Evaluation and Interpretability in Predictive Studies (Theme 4)	New. Covers missing data and KNN imputation, data leakage, evaluation on imbalanced data and SHAP interpretability, so that the literature also serves empirical questions RQ1 to RQ3. Draws on the six new verified sources together with Sylvain et al., Hirko et al. and Abujaber et al.
None	3.3.5 Methodological Review	New. Compares case study designs, broader quantitative studies and machine learning studies, with strengths and weaknesses of each.
None	3.4 Synthesis of the Literature	New. States what is known, where the literature disagrees and what remains unknown.
None (promised in the old introduction but never delivered)	3.5 Research Gap	New. States what is missing, why it matters and how the study fills the gap, using the template sentence pattern ("Despite extensive research on ..., little attention has been given to ...").
3.5 How will the literature guide my study? with 3.5.1 Guiding the Technical Design and Algorithm Selection	3.6 How the Literature Guides the Study	Heading made formal (no question, no first person). Expanded from algorithm selection only to also cover data preparation (20% missing data threshold, KNN with five neighbours, multiyear features) and interpretability.
3.6 Conclusion	3.7 Conclusion	Restates the three main issues, links the review explicitly to both theoretical objectives, explains how it informs the methodology and introduces Chapter 4.

Table 3: Comparison against the literature review tips

Tip from the images	Before	After
Introduction: topic, purpose, scope, organisation	Topic, purpose and outline present; scope (inclusions and exclusions) and organising approach not stated	All four elements present
Theoretical or conceptual background	Absent; concepts only in Chapter 1	Section 3.2 with conceptual framework
Thematic review of empirical studies	Topics present but not framed as themes	Four explicit themes with a stated logic
Methodological review	Absent	Section 3.3.5
Step 1: Start with a theme sentence	Some paragraphs open with a clear main idea; others open with background detail	Every paragraph opens with a theme sentence
Step 2: Introduce two or three authors	Usually present	Present in each main paragraph, mostly as narrative citations (for example "Seema et al. (2025:5) found ...")
Step 3: Show agreement or debate	Mostly summary; one implicit debate (Midvaal)	Explicit comparisons using "whereas", "while" and "similarly" in each theme
Step 4: Add your own voice	Mainly in end of section conclusions	Each paragraph states what the finding means for this study
Step 5: Link to the next theme	Sections ended without transitions	Link paragraphs at the end of Themes 1 and 2
Synthesis: connect the dots	Absent as a section	Section 3.4
Research gap	Mentioned in the introduction but never stated	Section 3.5
Summary and conclusion linked to objectives and methodology	Summary present; no link to objectives	Linked to both theoretical objectives and to the methodology

Table 4: In text citations added

All citations below were already used elsewhere in the report. None is a new source except that Obunga et al. and Phumlani et al. had not previously been cited in Chapter 3.

Citation	Used in new section	Taken from in the report
Phumlani et al., 2025:116	3.1	Chapter 1, 1.1 Introduction
Obunga et al., 2025:732	3.1, 3.2.3	Chapter 1, Binary Classification with Supervised Machine Learning
Ishola et al., 2025:76	3.2.1	Chapter 1, 1.2.1 and Chapter 5, 5.1
Ishola et al., 2025:81	3.2.1	Chapter 1, 1.2.1

Ishola et al., 2025:80	3.2.3, 3.6	Chapter 1, K Nearest Neighbours Imputation
Mamokhere, 2025:38	3.2.1	Chapter 1, 1.2.1
Nodipha et al., 2025:1372	3.2.1	Chapter 1, 1.2.1
Seema et al., 2025:4	3.2.1	Chapter 1, 1.2.1
S Mnguni, 2022:154	3.2.2	Chapter 1, The Gap in Compliance Performance and Audit Outcomes
S Mnguni, 2022:156	3.2.2	Chapter 1, The Gap in Compliance Performance and Audit Outcomes
Sambo, 2019:201	3.2.2	Chapter 1, The Gap in Compliance Performance and Audit Outcomes
Khasodi et al., 2023:347	3.3.2, 3.4, 3.7	The paper's conclusion on page 347, checked against the full text on 1 November 2026. Replaces page 338 for the Midvaal finding. (Page 340 was also considered but does not discuss clean audits alongside poor service, so it is not used.)
S Mnguni, 2022:153	3.3.5	Chapter 1, 1.1 Introduction
Hirko et al., 2026:29	3.2.3	Chapter 1, SHapley Additive exPlanations
Sylvain et al., 2025:3	3.3.4, 3.3.5, 3.6	Chapter 4, 4.3.6 Handling Missing Values Using KNN Imputation
Seema et al., 2025 (no page)	3.3.1	**New attribution.** The MMLM 70% capacity claim had no citation before; it is about MMLM, so it was attributed to Seema et al. Page number unknown.
Ishola et al., 2025:84 (wider use)	3.3.1	**Extended attribution.** Already cited; now also covers the national access figures (85,1% to 88,5%) and the urban and rural satisfaction figures (79% and 64%), which had no citation before.
Breiman, 2001:5; Troyanskaya et al., 2001:520; Saito & Rehmsmeier, 2015:1; Chicco & Jurman, 2020:1; Kapoor & Narayanan, 2023:1; Lundberg & Lee, 2017:4765	3.2.3, 3.3.3, 3.3.4, 3.3.5, 3.4, 3.6, 3.7	**New verified sources**, not previously in the report. Full details, the pages cited and every place each is used are in the separate New Sources Register PDF.

No citations were removed. The reference to the Constitution (1996) is still mentioned, as legislation, in 3.2.1.

Table 5: Content removed or corrected

Item	Before	After	Reason
Midvaal finding (Khasodi et al., 2023)	"Midvaal's consistent clean audits are a trustworthy sign of efficient service delivery" (old); the first rewrite kept this and presented Khasodi et al. as disagreeing with Sambo and Craig	Even Midvaal's clean audits "cannot be used as an effective determinant of good service delivery", but its internal controls position	The paper's conclusion (page 347) says the opposite of the old wording. The

		it to deliver services and produce clean audits (Khasodi et al., 2023:347)	synthesis (3.4) now presents the real debate: operational context (Craig) compared with internal controls (Khasodi et al.) and institutional capacity (Mamokhere)
City of Johannesburg and Water Infrastructure Quality Index (WIQI) sentence	"The results of the Water Infrastructure Quality Index (WIQI) which indicate that even cities ... such as the City of Johannesburg, nonetheless have problems ..."	Removed	Incomplete sentence with no citation
Stretch targets (Craig)	Departments "set lower performance targets, also known as stretch targets"	Departments "set modest performance targets rather than ambitious stretch targets"	Stretch targets are ambitious targets, so the old wording reversed the meaning
Who attains compliance	"communities may attain financial compliance"	"municipalities may achieve compliance"	Municipalities, not communities, are audited
Conclusion sentence on the AGSA	"the Auditor-General of South Africa's are not a true indicator of the actual water service delivery audit results"	"Audit outcomes are retrospective and remain a contested proxy for service delivery"	Old sentence was grammatically incomplete
Typing error	"an improved Random Forest mode"	"an improved Random Forest model"	Typing error
Section 3.5 heading	"How will the literature guide my study?"	"How the Literature Guides the Study"	Formal academic register, no first person question
Hyphenated proper names and terms	Modimolle-Mookgophong, Auditor-General, K-Nearest, Long Short-Term Memory, Sub-Saharan	Modimolle Mookgophong, Auditor General, K Nearest, Long Short Term Memory, Sub Saharan	Instruction to use no dashes of any kind

Table 6: Changes made outside Chapter 3

Change	Detail
Chapter 1, 1.1 Midvaal sentence	Now reads: "Even in the Midvaal Local Municipality, which has achieved consecutive clean audits, the audit outcome on its own cannot be used as an effective determinant of good service delivery (Khasodi et al., 2023:347)." The compliance gap section's citation was moved from page 340 to page 338, which supports it.
Chapter 1, research questions	Main question now refers to "publicly available municipal financial datasets". Added theoretical questions TRQ1 and TRQ2 to match the theoretical objectives. RQ1 split into a single clear question about the data source and KNN imputation.

Chapter 5, 5.2.1 and 5.2.2	Added answers for TRQ1 and TRQ2. Corrected the RQ2 results to match Chapter 4 (accuracy 91.7%, precision 91.8%, recall 96.3%, F1 0.940, ROC AUC 0.967). Corrected the RQ3 risk factors to the SHAP results in Chapter 4 (municipality identity, multiyear irregular expenditure trends, financial year). Corrected the objectives table: section numbers, fixed hyperparameters with 5 fold cross validation instead of a grid search, and the Chapter 4 metrics and SHAP findings.
References	Six new verified sources added in NWU Harvard format.
Title, abstract and Chapters 1, 2 and 5 aligned	New title: "Predicting Municipal Financial Governance Risk as an Early Warning Indicator for Water Service Delivery: A Machine Learning Approach". The abstract, aim, main research question, RQ2, RQ3, E OBJ2 to E OBJ4, scope, limitations, risk matrix, Chapter 2 evaluation design and all Chapter 5 answers, objectives evidence, contributions, limitations and conclusions now report the temporal results and no longer claim accurate prediction of water service delivery. E OBJ4 is marked "Partly achieved". Chapter 3 Sections 3.2.1, 3.2.3, 3.2.4, 3.3.4, 3.5 and 3.7 now describe financial governance risk as the predicted outcome.
Chapter 4 rewrite	Sections 4.1, 4.2, 4.3.7 and 4.4 to 4.7 rewritten to report the new temporal models: the financial governance risk target, previous year features, the temporal holdout and rolling backtest, both baselines, balanced accuracy and specificity, and a new SHAP analysis. Old Table 2 and Figures 8 to 11 replaced by Tables 2 to 4 and Figures 8 to 18 from thesis_figures. Chapter 3 Sections 3.3.4 and 3.6 adjusted so they no longer mention stratified sampling.
Imputation (Abstract, Chapters 1 to 5)	Claims that KNN imputation with five neighbours was used, and that variables with more than 20% missing values were dropped, were replaced with what the code actually does: records with unknown labels are excluded, and remaining missing values are filled with training year medians plus missing value indicators. Section 4.3.6 was rewritten, the Euclidean distance formula removed, and E OBJ1, RQ1, the risk matrix, the Chapter 5 answers and the limitations updated. Chapter 3 now discusses KNN as literature and explains why median imputation was preferred.

Table 7: Items still to verify

Issue	Detail
S Mnguni reference entry	The pages cited (148 to 157) match the article by Mnguni, S. and Subban, M. (2022) in the African Journal of Public Affairs 13(1), which online indexes list as pages 144 to 172 (one citing paper gives 144 to 171). The reference list gives a single author and page "29". Confirm the page range on the journal's own page and correct the entry.
Ishola et al. (2025:84) statistics	Confirm that the national access figures and the 79% and 64% satisfaction figures come from this source and page.
Seema et al. (2025) page number	Add the page number for the MMLM 70% capacity claim.
Sambo page numbers	Cited as :201 and :5 for the same paper.
Obunga et al. page number	Cited at :732, but the reference list gives pages 1 to 17.
Citation fields	New Chapter 3 citations are plain text, not EndNote fields, so EndNote will not update them.
Conceptual framework diagram	Optional: the template recommends a diagram; 3.2.4 describes one in words.